

TAIS MOTA

MECHANICAL ENGINEERING SENIOR AT UNIVERSITY OF SOUTH FLORIDA



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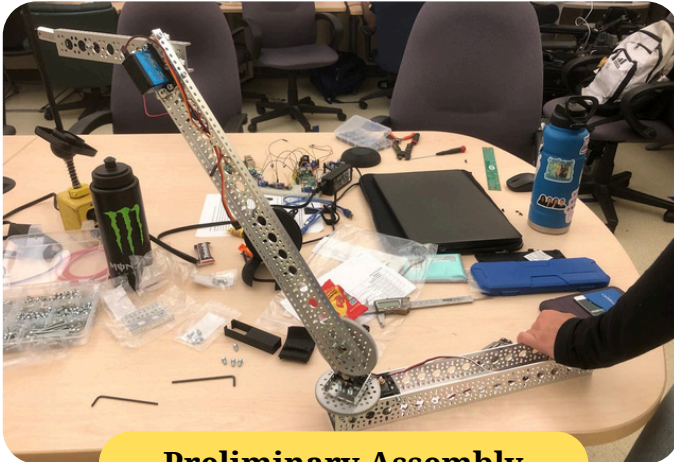


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1. ROBOTIC ARM - CARRT LAB



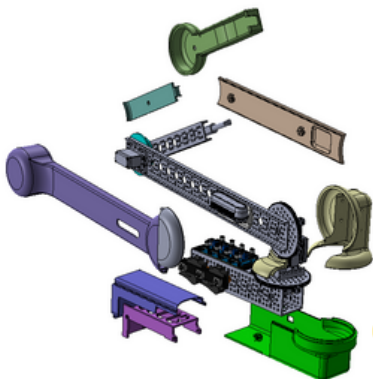
Preliminary Assembly

What?

- Robotic Arm that tackles pressing elevator buttons and card scanning.
- Simple, inclusive and affordable product for individuals with lower limb motor disabilities

How?

- Use **SolidWorks CAD** to design and prepare product for manufacturing.
- Integrated Arduino and Raspberry Pi, servo motors and others together as one functional unit
- Fabricated using various method such as **rapid prototyping (3D printing), machining, and soldering.**
- Design Printed Circuit Boards to minimize wiring and optimize electrical connections



Exploded CAD Model

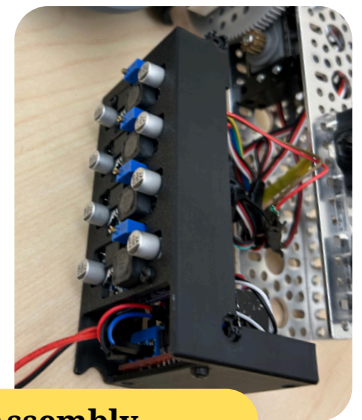
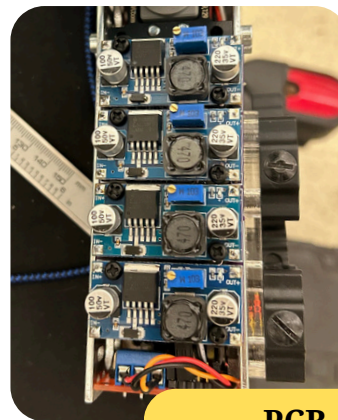
Key Features:

3 DOF
(Degrees of Freedom)

Retractable Resting
Position

Use User's
Wheelchairbattery

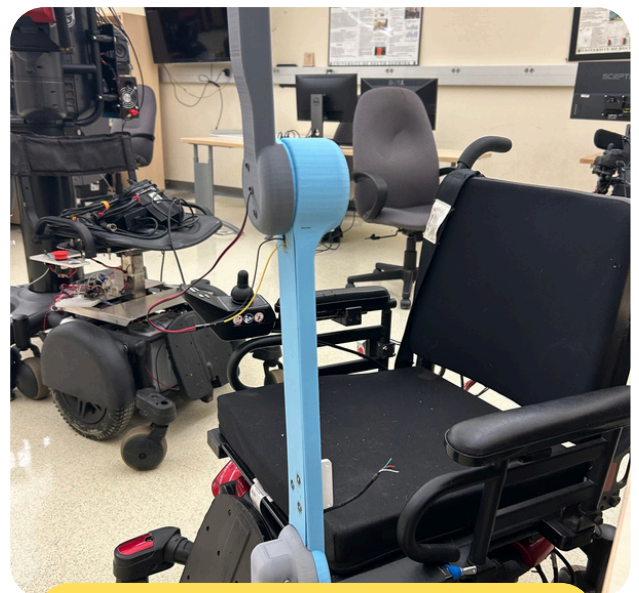
Various
End-Effector Options



PCB Assembly

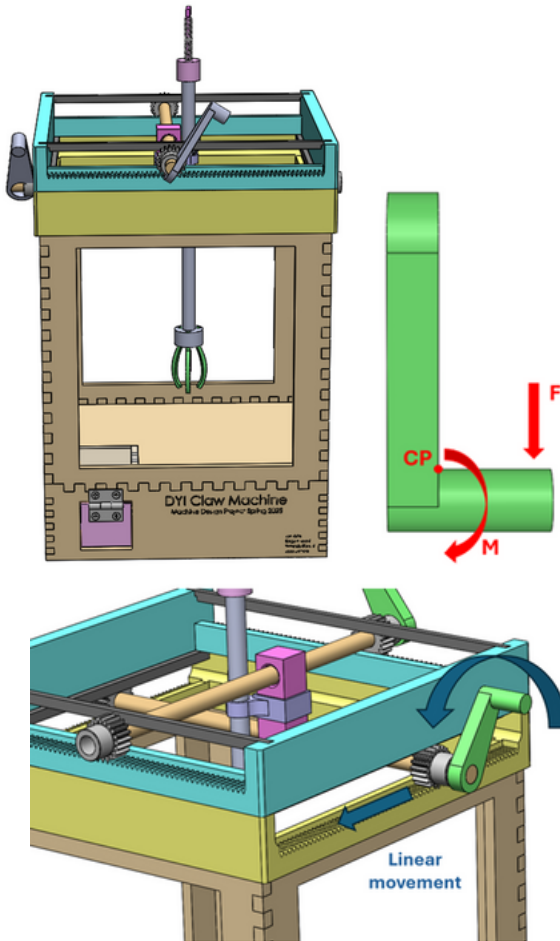
Results

- Provide product with optimized functionality and under market value by 20-40%
- Modular hardware system
- Future development of an automated system



Final Robotic Arm Assembly

2. DIY CLAW MACHINE



What?

Designed and built a purely mechanical claw machine with design requirements:

- Alter input velocity, torque/force.
- Incorporate at least: one gear, one linkage, one fastener, one shaft, and three standard components from McMaster-Carr/toolbox.
- CAD models, assembly, safety factor calculations.

How?

- Laser-cut plywood and acrylic panels frames
- Incorporated bolts, nuts, aluminum shafts, and gears from McMaster-Carr/toolbox; custom parts 3D printed in PLA.
- Conducted static and fatigue safety factor analysis in analytically and through MATLAB for shafts, cranks, hinges, and bolts.

Key Features:

Mechanical Design

Analytical Force Analysis

Rack-and-pinion gears and crank levers for motion conversion.

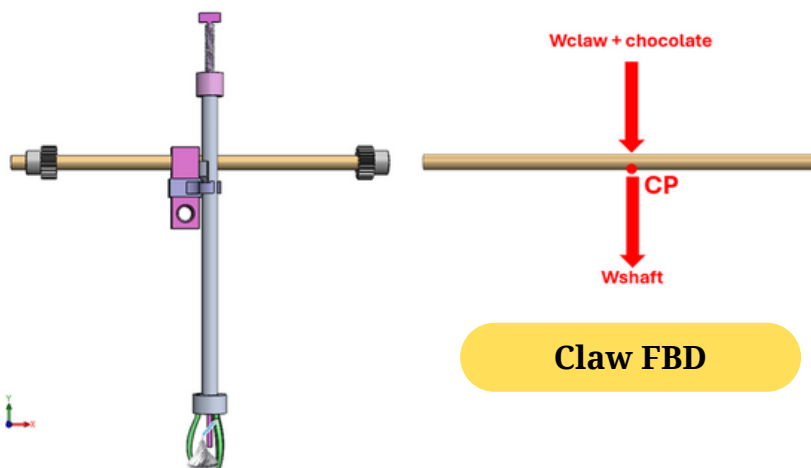
Spring-loaded claw with linkage-based release.

Combination of Laser-cut Wood and Acrylic, and 3D print

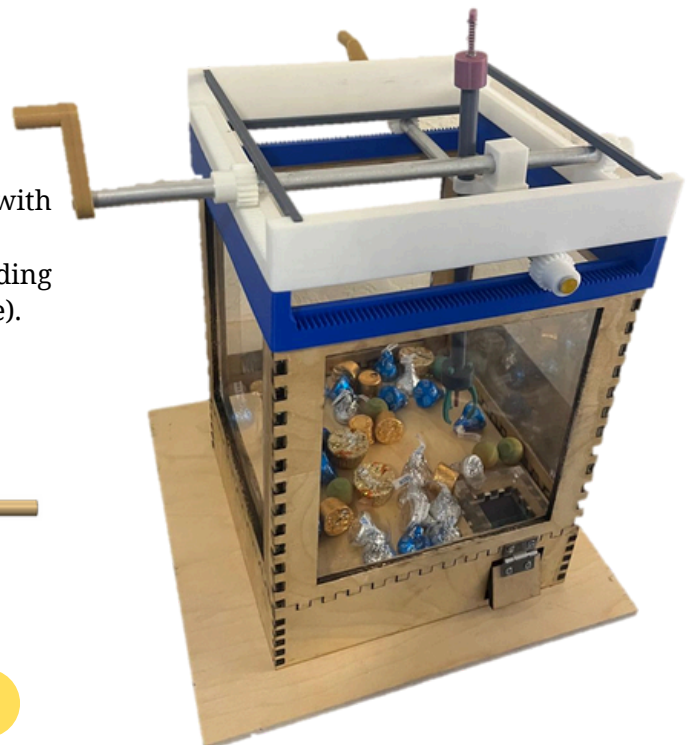
Geared Claw Control System

Results

- Delivered a fully functional, motor-free claw machine with smooth 3-axis control and reliable claw actuation.
- Verified high durability with calculated safety factors exceeding 1500 (static crank), 422 (fatigue crank), and >250 (shaft fatigue).



Claw FBD



Final Product Assembly

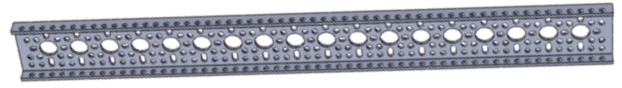
3. CHASSIS DESIGN ANALYSIS - SOLIDWORKS SIMULATION

What?

Assess in SolidWorks Simulation the bracket's performance:

- Evaluate Optimal Material under Static Loading Conditions
- Vibration Analysis
- Fatigue Life

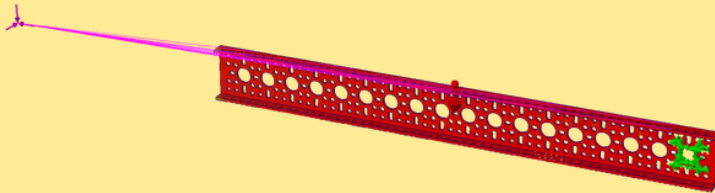
Goal: Achieve Factor of Safety of 2 across all tests



How?

1. Static Analysis for Material Selection

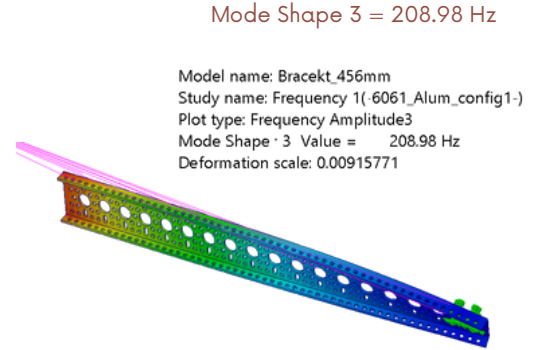
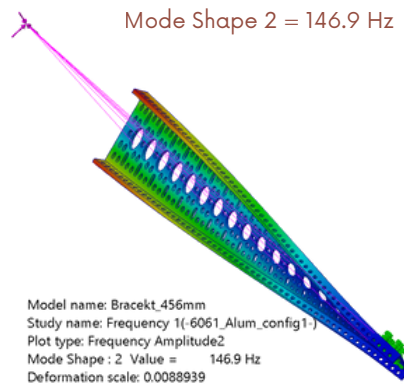
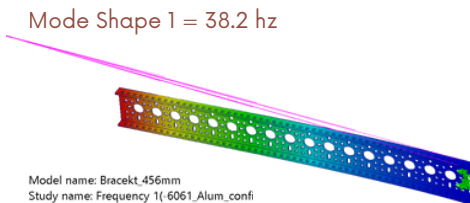
- Compared Brass, Alloy Steel, and 6061 Aluminum for weight, cost, and strength using Load Case Manager
- Applied mesh refinement and blended curvature meshing for accuracy



	Optimal (0)	Scenario 1	Scenario 2	Scenario 3
Model's Material	6061 Alloy	Alloy Steel	6061 Alloy	Brass
Minimum Factor of Safety (1)	4.576	45	4.576	17.67
Minimum Factor of Safety (2)	4.576	45	4.576	17.67
Mass 1	153.8 g	438.6 g	153.8 g	484.17 g

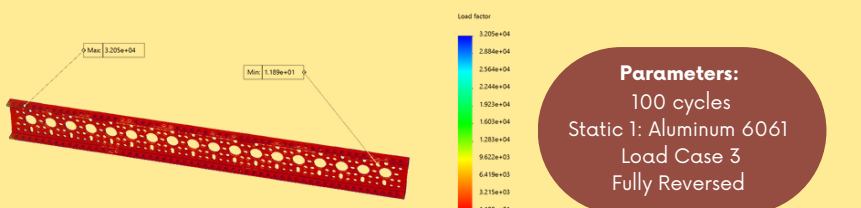
2. Vibration Analysis

- Verified first three modes at 38 Hz, 146 Hz, and 209 Hz
- Ensured all values remained well above operational range to minimize resonance risk



3. Fatigue Analysis

- Evaluated fatigue life under fully reversed cyclic loading
- Assessed stress-life curves to confirm durability of 6061 Aluminum
- Verified design longevity under repeated robotic actuation tasks



Key Results:

Validated bracket design as a reliable solution for robotic interaction tasks (e.g., elevator button actuation)

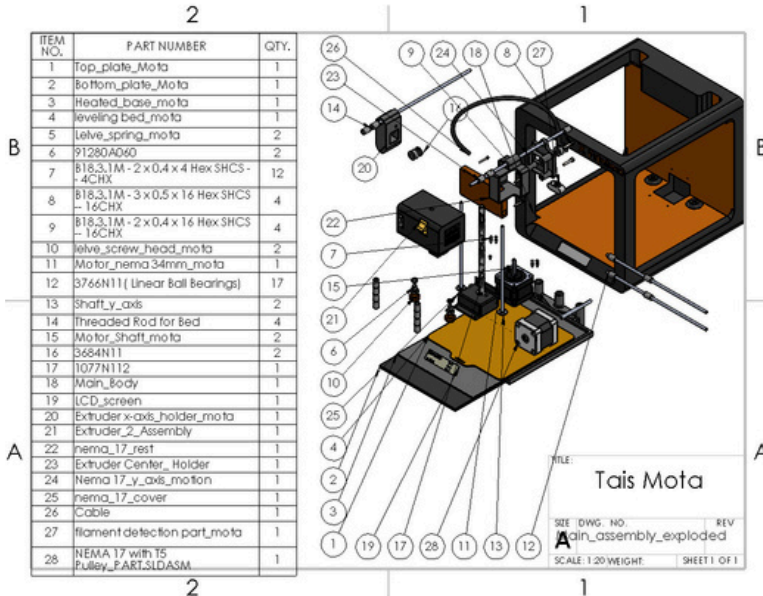
6061-T6 aluminum for strength, weight, and cost

Verified design with a factor of safety ≥ 2 for loadings

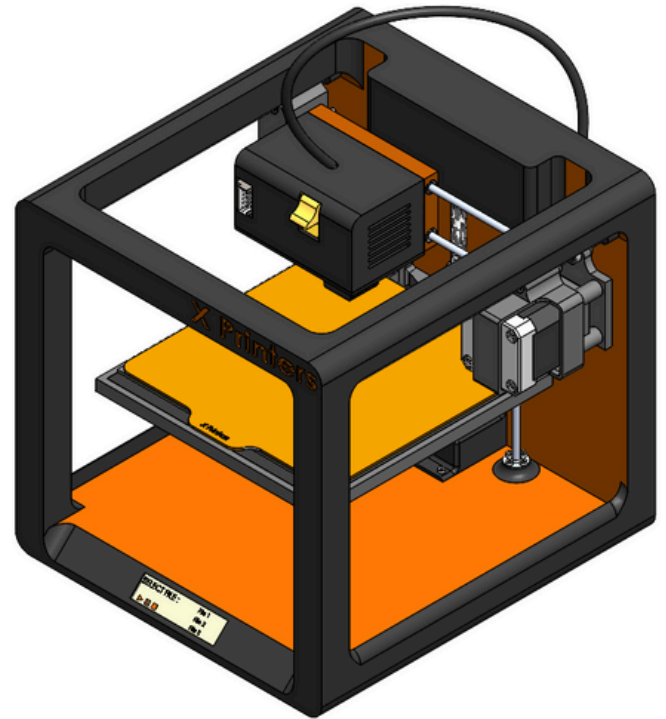
Applied mesh refinement and curvature-based meshing for improved accuracy

Natural frequencies (38 Hz, 146 Hz, 209 Hz) above operating range to avoid resonance

4. 3-D PRINTER CAD MODEL



Exploded View of 3D Printer



3D Printer CAD Model

What?

- Develop a functional mechanical system
- Develop Detailed Professional Drawings for each part designed

How?

- Used SolidWorks CAD's various features to produce a complete design
- Applied all steps of engineering product development cycle

Results

- Full CAD Model of 3D printer

5. NASA/METECS INERTIAL MEASUREMENT UNITS PROJECT

What?

- Build a reliable IMU protocol validated against motion capture for astronaut rehab tracking in any environment.
- Create a new VIS method enabling consistent data collection for astronaut exercise monitoring.



How?

- Conduct Data Collection in IMU and industry's standard (MOCAP) simultaneously
- Apply Inverse Kinematics tool using OpenSim software and DAS (Astronaut Simulation tool)
- Ensure data collection adheres to FDA regulation (CRF 312.62)



Vibration Isolation System (VIS)

IMU



Final Product Assembly

6. ML MODEL FOR CONTROLLING ACTIVE ASSISTIVE DEVICES USING EEG

What?

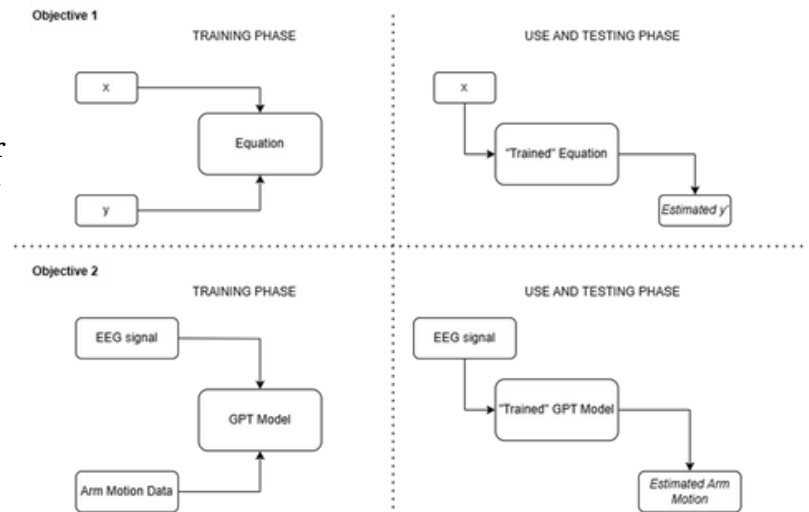
- Investigate the feasibility of leveraging a ChatGPT-based model to enable non-invasive brain-computer interface (BCI) control of assistive devices using EEG signals and synchronized motion-capture data

How?

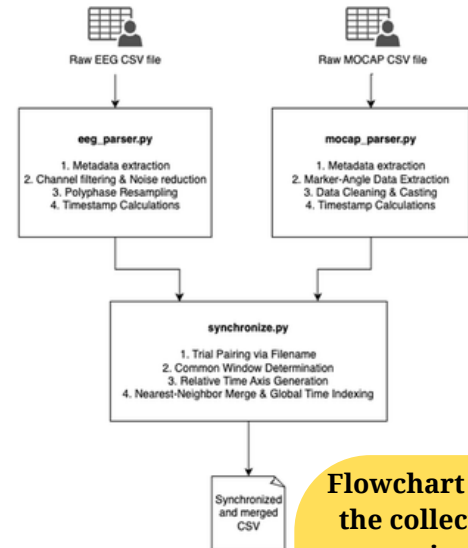
- Data Integration & Signal Processing:** Engineered a workflow to clean and synchronize noisy EEG signals with high-fidelity motion capture data, overcoming challenges in variability, sampling rate alignment, and artifact removal.
- System-Level Problem Solving:** Designed the pipeline with scalability and real-time application in mind, leveraging ChatGPT to rapidly prototype code while applying engineering rigor to ensure reproducibility, accuracy, and robustness

Results

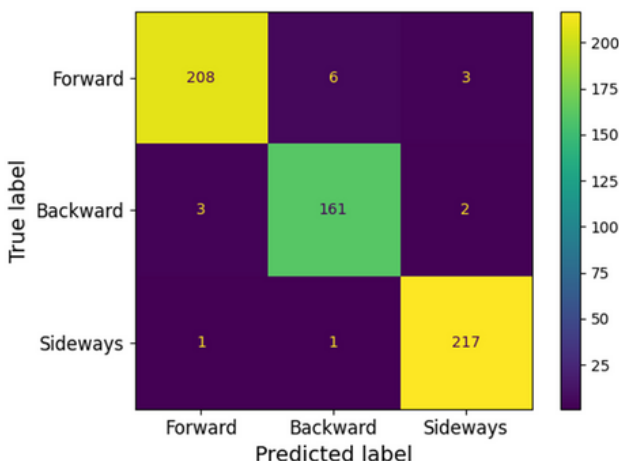
- Achieved **83.7%** mean accuracy with high-end EEG headset and **79.8%** accuracy with low-end headset → significantly higher than prior research (72% / 45%)
- CNN-LSTM regression predicted joint angles with moderate precision (MAE: Shoulder ≈ 22°, Elbow ≈ 5°, Wrist ≈ 16°)
- Demonstrated feasibility of combining EEG, MoCap, and GPT-driven pipelines for non-invasive assistive device control



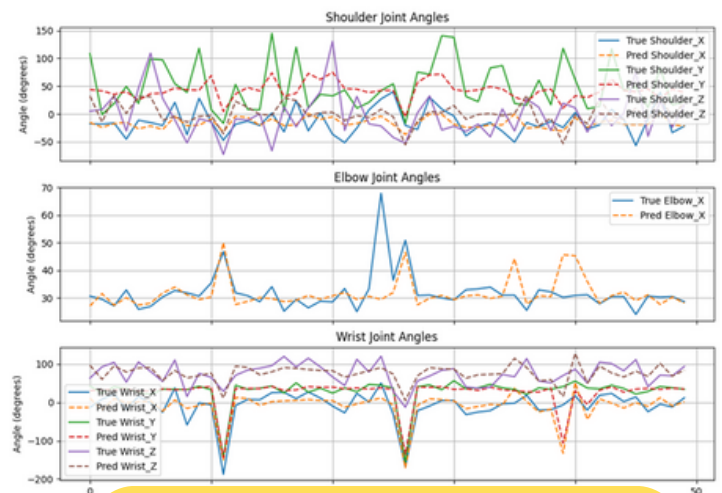
Model Functionality



Flowchart depicting the collective data processing required prior to inputting data into ChatGPT-4o



Confusion matrix of direction classification



Predicted vs. actual joint angles for shoulder, elbow, and wrist